Summary of Site Condition

Date Completed: 11/23/2012
MM/DD/YYYY

Site ID: 13437

Version 2.2

Purpose of this Summary of Site Condition

This Summary of Site Condition will serve several purposes. It will provide the Ministry of Environment (the ministry) with a summary of key information that will be used to understand the status of investigations and remediation, the nature and extent of remediation that is proposed or has been undertaken, further work that will be required, or closure documentation requested that is authorized by legislation and regulations in B.C. The Summary of Site Condition will also provide information to persons with an interest in investigations and management of contaminants on or adjacent to a property or properties that are considered a site.

The information contained in this Summary of Site Condition is provided by or on behalf of the British Columbia Ministry of Environment to assist individuals to become familiar with conditions and issues at a site for which contaminant investigations and / or remediation has been carried out and reviewed under the guidance of the British Columbia Contaminated Sites Regulation (CSR), the Hazardous Waste Regulation (HWR), and the *Environmental Management Act (*EMA*)*.

It is emphasized that this is a summary only and should in no cases, be the sole basis for important decisions about the site. Those with an interest in contaminant issues and the status of the site should seek more complete technical information as contained in site investigation, risk assessment, remediation plan and confirmation of remediation reports prepared by and signed by appropriately qualified individuals. Firms and individuals that rely on the information contained herein do so entirely at their own risk.

Notes and instructions

A Summary of Site Condition is to be completed by the Approved Professional(s) making submission to the Ministry of Environment with application for a regulatory instrument (e.g. Determination, Approval in Principle, Contaminated Soil Relocation Agreement or Certificate of Compliance).

This Summary of Site Condition will provide ministry regulatory officials with much of the information on which they will evaluate the recommendation of an Approved Professional(s).

A separate Summary of Site Condition is required for each service request submitted for a site.

All applicable parts of this Summary of Site Condition and required attachments (e.g., site plan; site plan showing areas of potential environmental concern, and / or areas of environmental concern) must be completed and submitted or it will be returned and processing of any application(s) will be delayed.

If the Summary of Site Condition is to accompany a recommendation by an Approved Professional that a service be provided as described in section 7.1 of the Contaminated Sites Regulation, the following must also be submitted with the package:

☒ a completed Contaminated Sites Service Application form
☒ a contaminated sites legal instrument cover letter (hard copy and electronic version)
☒ a completed draft contaminated sites legal instrument
☒ the applicable fees
☒ a signed Summary of Site Condition (hard copy and electronic version with PDF format preferred)

Failure to accurately fill out the Summary of Site Condition may result in delays issuing the legal instrument.

Part 1: Cover Page

(To be completed by the Approved Professional)

Current Site Owner: (*Attach additional sheets with names and contact information for additional site owners as required*)	**Mailing Address:** Company Name: Western Forest Products Inc., Inc. No. A0070374 Address: Suite 510 – 700 West Georgia Street P.O. Box 10032 City: Vancouver Postal Code: V7Y 1A1 Contact Name: J.M. (Jim) Stephen Phone: 604 665-6220 Fax: 604 682-2444 E-mail: jstephen@westernforest.com
Applicant *(If instrument is being applied for)* ☒ Same as above, or:	**Mailing Address:** Company Name: Address: City: Postal Code: Contact Name: Phone: Fax: E-mail:
Agent ☐ Same as applicant above, or:	**Mailing Address:** Company Name: Keystone Environmental Ltd. Address: Suite 320 – 4400 Dominion Street City: Burnaby Postal Code: V5G 4G3 Contact Name: Michael Geraghty Phone: 604 430-0671 Fax: 604 430-0672 E-mail: mgeraghty@keystoneenviro.com
Approved Professional(s) (*If making a recommendation under the CSR or another submission*)	**Mailing Address:** Company Name: Keystone Environmental Ltd. Address: Suite 320 – 4400 Dominion Street City: Burnaby Postal Code: V5G 4G3 Approved Professional Name: Lori C. Larsen, P.Ag. Phone: 604 430-0671 Fax: 604 430-0672 E-mail: llarsen@keystoneenviro.com Scope of review completed by Approved Professional: Standards Review of PSI 1, PSI 2, DSI, Confirmation of Remediation, Company Name: Steer Environmental Associates Address: 1515 Holland Street City: Nelson Postal Code: V1L 3E2 Approved Professional Name: Scott Steer Phone: 250-551-3507 Fax: none E-mail: scott.steer@steerenvironmental.com Scope of review completed by Approved Professional: Risk Assessment

Reason for Completing this Summary	**Role of Approved Professional:**
☒ Recommendation is being made, or: ☐ This is a submission without a recommendation under the CSR:	Reviews: ☒ Stage 1 preliminary site investigation report (Stage 1 PSI) ☒ Stage 2 preliminary site investigation report (Stage 2 PSI) ☒ Detailed site investigation report (DSI) ☐ Background substance concentrations report ☐ Remediation plan without risk assessment report ☐ Remediation plan with risk assessment report ☒ Confirmation of remediation report (CoR) ☒ Quantitative human health or ecological risk assessment report ☐ Screening level risk assessment report ☐ Other *(please specify)* Recommendation(s) *(With Regulatory Instrument)*: ☐ Determination *(Determination)* ☐ Approval in Principle, numerical standards *AiP numerical standards)* ☐ Approval in Principle, risk-based standards *(AiP risk-based standards)* ☐ Contaminated Soil Relocation Agreement *(CSRA)* ☐ Certificate of Compliance, numerical standards *(CoC numerical standards)* ☒ Certificate of Compliance, risk-based standards *(CoC risk-based standards)* ☐ Other *(please specify)* Section 4 of the Summary of Site Condition does not need to be completed with the request for Certificate of Compliance where an Approval in Principle exists for the site provided that no new information has been obtained for the site applicable to this section of the form.

Part 2: Executive Summary

(To be completed by the Approved Professional(s) reviewing site investigation, risk assessment, remediation or confirmation of remediation reports)

Site Location: *(The site and location plans are to be provided as Schedule A of the draft instrument.)*	
Subject Site:	
Civic Address(s):	None
Site Common Name: *(if applicable)*	Jordan River Group 2 (JRG2)
Legal description(s) *or* metes and bounds: *(add additional pages if needed)*	Parcel B (DD 99860I) of Section 9, Renfrew District, Except those parts in Plans 13455, 32802 AND VIP58390
PID(s): *(or PIN(s) if untitled Crown land)*	009-574-212
Civic Address(s):	None
Site Common Name: *(if applicable)*	Jordan River Group 3 (JRG3)
Legal description(s) *or* metes and bounds: *(add additional pages if needed)*	Section 9, Renfrew District, Except Parcel A (DD 98685I), Parcel B (DD 99860I), and Parcel C (DD 170967I) and except part in Plan VIP58390
PID(s): *(or PIN(s) if untitled Crown land)*	000-643-220
Centre of Entire Site: *(using NAD 83 convention)* *(accurate to ± 0.5 second)*	Latitude: 48 degrees 25 min 27.5 secs
	Longitude: 124 degrees 2 min 46.5 secs
Offsite impacted Properties *or* Receiving Site:	☐ Offsite impacted propertie(s) – provide information for each ☐ Receiving site for Contaminated Soil Relocation Agreement ☐ Not Applicable
Civic Address(s):	
Site Common Name: *(if applicable)*	
Legal description(s) *or* metes and bounds (if a portion of a site): *(add additional pages if needed)*	
PID(s): *(or PIN(s) if untitled Crown land)*	
BC Site ID *(if applicable)*:	
Approximate Centre of site:	Latitude: ____ degrees ____ min ____ secs
(accurate to ± 0.5 second)	Longitude: ____ degrees ____ min ____ secs

Part 3: Document Summary

(List of all known site investigation, risk assessment (including screening level risk assessment), remediation plan and confirmation of remediation reports completed and directly supporting correspondence submitted (subject site and offsite impacted sites).

#	Document Title	Author / Company	Document Date
1	Report of Findings: Human Health and Ecological Risk Assessment Jordan River Group Properties 2 and 3 Jordan River, BC	Keystone Environmental Ltd.	November 2012
2	Report of Findings: Stage 1 Preliminary Site Investigation Update, Stage 2 Preliminary Site Investigation, Detailed Site Investigation and Confirmation of Remediation - Jordan River Group Properties JRG 2 and JRG3, Jordan River, BC	Keystone Environmental Ltd.	June 2012
3	Phase I Environmental Site Investigation, Jordan River Group, Southern Vancouver Island WFP Lands, Jordan River, BC.	Golder Associates Ltd.	March 24, 2008
4	Hydrogeologic Assessment and Water Quality Monitoring of the Wood Waste Landfill Site, Jordan River, British Columbia	Golder Associates Ltd	December 12, 2007

Part 4: Investigation Summary

4.1 Investigations Completed

		Yes	No	n/a
Stage 1 PSI	Completed?	☒	☐	☐
	Includes Stage 1 PSI information as listed in CSR S.58 and any current applicable ministry protocols, guidelines, checklists, etc.?	☒	☐	☐
Stage 2 PSI	Completed?	☒	☐	☐
	Includes Stage 2 PSI information as listed in CSR S.58 and any current applicable ministry protocols, guidelines, checklists, etc.?	☒	☐	☐
DSI	Completed?	☒	☐	☐
	Includes DSI information as listed in CSR S.59 and any current applicable ministry protocols, guidelines, checklists, etc.?	☒	☐	☐
Other Reports	Completed? *(Specify in Notes below)*	☒	☐	☐
	According to other guidelines? *(Provide explanation in notes below. Indicate how reports assist understanding of conditions and remediation.)*	☒	☐	☐
Notes: **Other Reports**: Confirmation of Remediation – Following current applicable ministry protocols and guidelines				

n/a – not applicable
If completed investigation reports are not adequate or if reports are titled differently or have a different scope than those listed above in accordance with the Contaminated Sites Regulation (i.e., PSI, DSI), complete Section 4.8 (Investigation or Interpretation Issues).

4.2 Site Conditions

Topography

Describe steepness and direction of slope and position of site in relation to surrounding land

The Site is divided into two distinct geographies: a sea level low lying area; and high lands area. The difference in ground elevation between the two areas is approximately 35 m.

The topography of JRG2 is generally flat, sloping gently to the ocean to the south and Jordan River to the west. However the northern edge of the property begins a steep rise to the north in the area of the wood waste landfill.

The topography of JRG 3 is similar to JRG2 in the southwest portion however the land rises steeply in the northern and eastern portions of the property.

Stratigraphy

Describe depth and thickness, grain size, etc. of typical stratigraphic components and note depth to cemented or very compact materials, bedrock / refusal, etc.

The village of Jordan River is located on a deltaic outwash plain of the Jordan River over laying glacial and post-glacial sands, gravels, and clays

The unconsolidated stratigraphy encountered at the Site during the investigations consisted of two distinct zones; 1) fill material, and 2) undisturbed native soils. Portions of the delta were at one time more swamp like, as indicated by the organic rich silts beneath the dryland sort, anecdotal stories, and historical photographs. These areas have been filled over the years with local material to make habitable and workable land.

JRG 3 North East Upland Area - Former Wood Waste Landfill and Shake Mill Area (excluding MW08-17 and MW08-18)

Unit	Approximate Depth of Subsurface Layer	Description
A	From the surface to approximately 1.5 m bg	brown, medium grained sand, sand and gravel FILL, loose to medium dense, dry to moist, overlies Unit B (wood waste) at MW08-83, MW11-105, MW11-106, and BH11-113.
B	From approximately 1.5 mbg to 4.6 m bg	Light brown WOOD WASTE, loose, moist to wet. At some locations organic odours were noted. Staining was not observed. Present at MW08-39, MW08-83, MW11-105, MW11-106, BH11-113
C	From the surface (where A and B not present) to between approximately 1.5 m bg to 5.6 mbg.	Brown SAND and GRAVEL, loose, dry to wet. Odours not observed. Mottled staining at MW08-19.
D	From the surface (where A and B not present) to approximately 7.6 m bg	Brown SAND, loose, dry to moist, no odours
E	From approximately 5.6 m bg to 6.0 m bg and 16 m bg to 18 m bg.	Grey/light brown sandy SILT, medium dense, wet. Odours and/or staining not observed.
F	From between approximately 3.0 mbg to 5.6 mbg, to a depth of between approximately 4.5 mbg to 7.5 mbg	Brown to grey, fine to coarse grained SAND or silty SAND, loose, dry to wet. Odours and/or staining not observed.

Transitional Slope Between JRG3 Upland to Lowland Area - MW08-17 and MW08-18: At the two monitoring wells located closest to the southern escarpment, MW08-17 and MW08-18, the stratigraphy consisted of brown sand and gravel, which was loose in compactness, and dry to wet ranging from surface to approximately 16 m bgs. This unit was underlain by saturated, dense, light brown silt with some sand that ranged from 16 m bgs to the final drill depth of 18 m bgs.

Deeper Lithology: Using the lithology information recorded on the Detailed Well Records of tag numbers 81862, 101523, and 93495, the deeper stratigraphy consisted of:

- GRAVEL and COBBLES from surface to 6.7 m
- SAND and GRAVEL, lenses of silty SAND from 6.7 m to 14.6 m
- BEDROCK starting at 14.6 m
- conglomerate BEDROCK from 21 m to 46 m
- sandstone BEDROCK from 46 m to 56 m
- interbedded conglomerate and sandstone BEDROCK 56 m to 104 m
- hard green BEDROCK at 104 m

JRG 2 and 3 Lowland Areas

Unit	Approximate depth of subsurface layer	Description
A	From the surface to a depth of between approximately 0.4 mbg to 4.5 mbg.	Brown, dark brown, to grey SAND and GRAVEL, sandy GRAVEL, loose, dry to wet. Odours and/or staining not observed.
B*	From the surface to a depth of approximately 1.8 mbg to 3.0 mbg	Brown to grey sand or SILT (FILL), loose to medium dense, dry to moist. Odours and/or staining not observed.
C	From between approximately 0.4 mbg to 4.5mbg	Grey to brown, fine to coarse grained SAND, trace to minor gravel, some interbedded lenses of SILT to sandy SILT, minor organics, loose to medium dense, dry to wet. At some locations organic odours were noted. Staining not observed.
D**	approximately 1 mbg to 4.6 mbg	Blue grey CLAY, soft to very dense, wet. No odour or staining.

Deeper Lithology of Lowland Areas
Using the lithology information recorded on the Detailed Well Records of tag numbers 65106, 65107, and 65108, the deeper stratigraphy consisted of:

- GRAVEL and silty GRAVEL from surface to 9.5 m
- silty SAND, and SAND from 7.6 m to 29 m
- silty/clay gravel TILL from 17 m to 70 m or a blue CLAY from 27 m to 30 m
- bedrock at 70 m

Hydrogeology

Describe groundwater levels, confining / semi-confining layers, flow direction and velocity

In general, groundwater flow direction is towards the Strait of Juan de Fuca or the Jordan River, although localized influences such as underground utilities (water pipes) and groundwater discharge to surface from the hillsides can affect the groundwater flow direction.

The main portion of the Site is situated on the Jordan River deltaic sediments, at a significantly lower elevation than the inland area at the northern portion of the Site and northward (approximately 35 m lower). The decrease in ground elevation is steep with numerous groundwater seeps along the slope face as the water table is significantly higher on the uplands than on the delta and foreshore. The water table is at depths ranging between 1.5 m and 3 m in the lowland area (maintenance shop area) and approximately 10 below the ground surface in the inland (upland) areas.

The Site and surrounding areas receive high volumes of precipitation mainly in the form of rain. Infiltrations rates are assumed to be high due to the porous nature of the soils, but overland flow on sloped areas is also anticipated to be high during storm events with some run off being directed into drainage ditches. Both cases are anticipated to contribute substantially to groundwater recharge.

Groundwater flows south and southwest until discharge into the ocean. The groundwater flow direction in the maintenance shop area appears to be influenced by recharge from ditch #2 as groundwater is shallower closest to the ditch and deeper in the south-western portion of the Site.

The on-site water well, located at the southern portion of the Site within the pump house, was inferred to be screened within a deep aquifer (44 m depth) based upon the detailed well record. Chemical analyses of the groundwater at this location indicated high concentrations of sodium (1,670 mg/L), magnesium (346 mg/L), chloride (3,900 mg/L), and TDS (6950) which are suggestive of salt water intrusion from ocean waters. Investigation monitoring wells did not contain similarly elevated values of sodium, magnesium or chloride, therefore it is anticipated that a salt-water wedge, if present, underlies a portion of the deltaic flats area at greater depths than the maximum depth of investigation (15 mbg). At depth, the saltwater wedge would likely be tidally influenced. As a confining unit or aquitard was not established between the top of the water table and the deeper drinking water aquifer, the Site is considered to contain one aquifer.

Vertical migration of fresh water would be limited by the salt water wedge, if present, resulting flow along the boundary between the fresh and salt waters.

Tidal influence on groundwater is typical of unconfined conditions with amplitude generally attenuated within 50 m to 75 m of the shoreline

Upland Area of JRG2:

Previous hydrogeological investigation by Golder in 2007 at the wood waste landfill area indicated that a shallow aquifer was "perched" above a native silt unit that restricts downward migration of groundwater. Golder inferred that the dominant groundwater flow path is with the shallow sand and gravel unit overlying the silt; groundwater monitoring indicated that the influence of tidal effect on the shallow groundwater aquifer was relatively minor. Groundwater flow direction from the landfill area is inferred to range from the south to the northwest (toward the Jordan River) and the seep location appears to be localized with relatively low flow (1.8 L/min to 17 L/min). An abandoned spillway is present to the south of the landfill area and collects seepage from the landfill.

Surface Water Features

List name, direction and distance to nearest surface water bodies and the characteristics (e.g., relative size / flow) of the water body

Fresh water:

Jordan River – Adjacent to Site to the west.

Marine waters:

Strait of Juan de Fuca, adjacent to the south of JRG2 and adjacent to the south of the east portion of JRG3

4.3 Land Use

Location		Description of Current Land Use(s) / Activities
Onsite	Subject site	Uplands – Forested **JRG2:** Former Dryland Sort Landfill (PR-6965) mostly on JRG2 and partially on JRG3. Spillway, created by Victoria Light and Power in 1915, runs east to west on the southern portion of JRG2. There are also two residential houses, one burnt residential house, and forested lands **JRG3:** Dryland Sort Landfill (PR 16140) on the east uplands on JRG3 Community of Jordan River on the lowlands of the east side of JRG3 – houses the Western Forest Products office and fuelling, storage, and maintenance areas.
Offsite	North	Forested Lands North of JRG2 and east of the west portion of JRG3 – Abandoned BC Hydro Jordan River Power House
	East	Forested Lands
	South	Ocean – Strait of Juan de Fuca
	West	Jordan River

Proposed land use of subject site: ☐ same as above or ☒ other (please specify)

Proposed parkland

4.4 Applicable Numerical Concentration Standards and Criteria

(if more than one land or water use applies to the site, expand this section to specify additional land uses covered by the instrument, i.e. riparian areas, roadways, etc. Include a diagram to clearly show the areas with different standards)

Soil (standards):

Property		CSR Land Use					
		AL	PL	RL	CL	IL	Other
Subject site	Current	☐	☐	☐	☒	☒	☐
	Proposed	☐	☒	☐	☐	☐	☐
Receiving site *(if completed in support of a Contaminated Soil Relocation Agreement)*		☐	☐	☐	☐	☐	☐
Offsite impacted property / management area		☐	☐	☐	☐	☐	☐

If Other is specified above, please explain: *(applicable or excluded guidance, protocols or policies specific to the site)*

Water (standards):
(Check all that apply)

	AW fresh	AW marine	IW	LW	DW	No Water Use
Groundwater (CSR Schedule 6 & 10)	☒	☒	☐	☐	☐	☒
	Ambient fresh	Ambient marine				
Surface Water (Ambient Guidelines and/or Criteria)	☒	☒	☐	☐	☐	

Sediment (criteria CSR Schedule 9):

Type of Aquatic Life		Type of habitat		
☐ Freshwater	☐ Marine/Estuarine	☐ Sensitive	☐ Typical	☒ Not applicable

4.5 APEC and PCOC Summary

(Not applicable for a receiving site in a Contaminated Soil Relocation Agreement)
Provide reference to a figure showing onsite and offsite areas of potential environmental concern (APEC) and contaminants of potential concern associated with each APEC:

Report # 2 Figure # 4 Page # n/a

Area of Potential Environmental Concern (APEC)		Potential Contaminant of Concern (PCOC) *(indicate products, chemicals, waste type, etc. and / or analytical parameter)*	Check where analyses completed					
#	**Description** *(describe location in relation to process source, waste, filling, land use or activity, etc. giving rise to APEC, and if APEC is primary due to soil or water contamination)*		Soil	Sediment	Ground water	Surface Water	Vapour	Other (explain)
1	**Area: JRG2 (part JRG3)** Wood waste landfill	Anions	☐	☐	☒	☐	☐	☐
		metals	☐	☐	☒	☐	☐	☐
		chemical oxygen demand (COD),	☐	☐	☒	☐	☐	☐
		biological oxygen demand (BOD)	☐	☐	☒	☐	☐	☐
		tannins/lignins	☐	☐	☒	☐	☐	☐
		phenols	☐	☐	☒	☐	☐	☐
		resin and fatty acids	☐	☐	☒	☐	☐	☐
		Sulphides and chlorides	☐	☐	☒	☐	☐	☐
2	**Area: JRG3** Former Building Sites – Burnt Foundations	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		Metals	☒	☐	☒	☐	☐	☐
			☐	☐	☐	☐	☐	☐

Area of Potential Environmental Concern (APEC)		Potential Contaminant of Concern (PCOC)	Check where analyses completed					
#	**Description** *(describe location in relation to process source, waste, filling, land use or activity, etc. giving rise to APEC, and if APEC is primary due to soil or water contamination)*	*(indicate products, chemicals, waste type, etc. and / or analytical parameter)*	Soil	Sediment	Ground water	Surface Water	Vapour	Other (explain)
3	**Area: JRG3** Wood Waste Landfill	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☒	☐
		VPHv	☐	☐	☐	☐	☒	☐
		MTBE	☐	☐	☐	☐	☒	☐
		VOC	☐	☐	☐	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐
		Anions	☐	☐	☒	☐	☐	☐
		chemical oxygen demand (COD),	☐	☐	☒	☐	☐	☐
		biological oxygen demand (BOD)	☐	☐	☒	☐	☐	☐
		tannins/lignins	☐	☐	☒	☐	☐	☐
		phenols	☒	☐	☒	☐	☐	☐
4	**Area: JRG3** Current Workshop – vehicle and equipment maintenance, waste oil storage	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		VOC	☒	☐	☒	☐	☒	☐
		MTBE	☐	☐	☐	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐
		Ethylene glycol	☒	☐	☐	☐	☐	☐
5	**Area: JRG3** Former Workshop – vehicle and equipment maintenance, waste oil storage	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		VOC	☒	☐	☒	☐	☒	☐
		MTBE	☐	☐	☐	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐
		Ethylene glycol	☒	☐	☒	☐	☐	☐

Area of Potential Environmental Concern (APEC)			Check where analyses completed					
#	**Description** *(describe location in relation to process source, waste, filling, land use or activity, etc. giving rise to APEC, and if APEC is primary due to soil or water contamination)*	**Potential Contaminant of Concern (PCOC)** *(indicate products, chemicals, waste type, etc. and / or analytical parameter)*	Soil	Sediment	Ground water	Surface Water	Vapour	Other (explain)
6	**Area: JRG3** Former Fuelling Station	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		VOC	☒	☐	☒	☐	☒	☐
		MTBE	☐	☐	☐	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐
7	**Area: JRG3** Chain Saw Fuel Shed	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		VOC	☒	☐	☒	☐	☒	☐
		MTBE	☐	☐	☐	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐
8	**Area: JRG3** Metal Scrap Shed – associated above ground storage tanks	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		VOC	☒	☐	☒	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐
		Ethylene glycol	☒	☐	☒	☐	☐	☐
9	**Area: JRG3** Burn Pit	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		Metals	☒	☐	☒	☐	☐	☐
10	**Area: JRG3** Former Oil House – small shed with wooden floor located NE of current maintenance shop used to store oil	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		VOC	☒	☐	☒	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐

Area of Potential Environmental Concern (APEC)		Potential Contaminant of Concern (PCOC)	Check where analyses completed					
#	**Description** *(describe location in relation to process source, waste, filling, land use or activity, etc. giving rise to APEC, and if APEC is primary due to soil or water contamination)*	*(indicate products, chemicals, waste type, etc. and / or analytical parameter)*	Soil	Sediment	Ground water	Surface Water	Vapour	Other (explain)
11	**Area: JRG3** Former Store/workshop – a store and shop located east of the current WFP Office. Equipment repairs were potentially conducted in the shop	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		VOC	☒	☐	☒	☐	☒	☐
		MTBE	☐	☐	☐	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐
		Ethylene glycol	☒	☐	☒	☐	☐	☐
12	Former Herbicide Storage Shed – small shed with wooden floor used to store moderate quantities of herbicides located east of the current WFP Office	Metals	☒	☐	☒	☐	☐	☐
		Chloro acid Herbicides	☒	☐	☒	☐	☐	☐
		Pesticides	☒	☐	☒	☐	☐	☐
13	Former Chainsaw Fuel Storage Shed located east of the current WFP Office	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☐	☐	☐	☐
		VPH	☒	☐	☐	☐	☐	☐
		VOC	☒	☐	☐	☐	☐	☐
		MTBE	☒	☐	☐	☐	☐	☐
		Metals	☒	☐	☒	☐	☐	☐
14	**Area: JRG3** Former Truck Wash – concrete pad with no O/W separator	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☐	☐
		VPH	☒	☐	☒	☐	☐	☐
		VOC	☒	☐	☒	☐	☐	☐
		MTBE	☒	☐	☒	☐	☐	☐
		Metals	☒	☐	☒	☐	☐	☐
		Ethylene glycol	☒	☐	☒	☐	☐	☐
15	**Area: JRG3** Former Shake Mill – Heating oil UST	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☒	☐
		VPH	☒	☐	☒	☐	☐	☐
		VPHv	☐	☐	☐	☐	☒	☐
		Naphthalene	☐	☐	☐	☐	☒	☐
		Metals	☒	☐	☒	☐	☐	☐

Area of Potential Environmental Concern (APEC)		Potential Contaminant of Concern (PCOC) (indicate products, chemicals, waste type, etc. and / or analytical parameter)	Check where analyses completed					
#	Description (describe location in relation to process source, waste, filling, land use or activity, etc. giving rise to APEC, and if APEC is primary due to soil or water contamination)		Soil	Sediment	Ground water	Surface Water	Vapour	Other (explain)
16	Area: JRG3 Current First Aid Building – Heating oil AST	LEPH	☒	☐	☒	☐	☐	☐
		HEPH	☒	☐	☒	☐	☐	☐
		PAH	☒	☐	☒	☐	☐	☐
		BTEX	☒	☐	☒	☐	☐	☐
		VPH	☒	☐	☒	☐	☐	☐
		Metals	☒	☐	☒	☐	☐	☐
			☐	☐	☐	☐	☐	☐

Other (please explain):

APEC #	

4.6 AEC and Contaminant Summary

Stage 2 PSI - Provide reference to figure(s) showing the areas of environmental concern (AEC) and contaminants of concern associated with each AEC in onsite and offsite soil, water, sediment and/or vapour. Sample locations and corresponding analytical results shall be shown on each figure and in tabular form with reference to applicable standards:

Environmental Medium	Soil	Report #	2	Figure #	6	Page #	n/a
Environmental Medium	Groundwater	Report #	2	Figure #	7	Page #	n/a
Environmental Medium	Soil Vapour	Report #	2	Figure #	8	Page #	n/a

DSI – Provide references to figures (plan and section), with contours, showing the specific lateral and vertical distribution of each contaminant of concern in onsite and offsite soil, sediment, water and vapour. Sections shall be longitudinal and transverse with respect to groundwater flow and include physical conditions (e.g. stratigraphy, water table etc.). Sample locations with corresponding analytical results used to develop each figure shall be shown on the figure and in tabular form with reference to applicable standards:

Environmental Medium	Soil	Report #	2	Figures #	6	Page #	n/a
Environmental Medium	Groundwater	Report #	2	Figures #	7	Page #	n/a
Environmental Medium	Soil Vapour	Report #	2	Figures #	8	Page #	n/a

AEC / APEC # (Use same #s as for APECs in Table above)	Contaminant of Concern	Medium (e.g., soil, groundwater, sediment, vapour, surface water, other)	Maximum Measured Concentration (indicate units)	Extent of Contamination	
				Area (m^2)	Depth Range (m)
1	sulphide	surface water	0.046 mg/L	15,000	n/a
1	chloride	surface water	10,000 mg/L	15,000	n/a
3	non-chlorinated phenols	groundwater	413 µg/L	450	n/a
3	HEPH	Soil	3,270 µg/L	330	1.5 – 2.3 m
4	LEPH	soil	1,730 µg/g	487	0.3 – 1.5m
4	HEPH	soil	1,780 µg/g	487	0.3 – 1.5m

AEC / APEC # (Use same #s as for APECs in Table above)	Contaminant of Concern	Medium (e.g., soil, groundwater, sediment, vapour, surface water, other)	Maximum Measured Concentration (indicate units)	Extent of Contamination	
				Area (m^2)	Depth Range (m)
4	lead	soil	223 µg/g	20	0 – 0.5 m
4	zinc	soil	201 µg/g	285	1.5 – 3.5 m
4	zinc	soil	186 µg/g	320	0.76 – 1.5 m
4	benzene	vapour	2.6 µg/m^3	384	n/a
5	HEPH	soil	1300 µg/g	318	0 – 1.5 m
5	copper	soil	185 µg/g	216	0 - 2.9 m
5	lead	soil	1,810 µg/g	206	0 - 2.9 m
5	tin	soil	625 µg/g	190	0.76 – 3.5 m
5	zinc	soil	898 µg/g	190	0.76 – 3.5 m
6	LEPH	soil	2,900 µg/g	316	0.76 – 2.6 m
6	LEPHw	groundwater	1,300 µg/L	190	2.5 – 7.1 m
6	naphthalene	groundwater	22 µg/L	190	2.5 – 7.1 m
6	HEPH	soil	1300 µg/g	316	0.76 – 2.6 m
6	benzo(a)anthracene	soil	2.5 µg/g	316	0.76 – 2.6 m
6	benzo(b)fluoranthene	soil	3.3 µg/g	316	0.76 – 2.6 m
6	benzo(k)fluoranthene	soil	1.1 µg/g	316	0.76 – 2.6 m
6	benzo(a)pyrene	soil	2.2 µg/g	316	0.76 – 2.6 m
6	indeno(1,2,3-cd)pyrene	soil	1.6 µg/g	316	0.76 – 2.6 m
10	LEPH	soil	7,200 µg/g	802	0.3 – 4.0 m
10	HEPH	soil	51,000 µg/g	802	0.3 – 4.0 m
10	ethylbenzene	soil	2.1 µg/g	640	0.3 – 4.0 m
10	xylenes	soil	20 µg/g	640	0.3 – 4.0 m
10	VPH	soil	1,700 µg/g	640	0.3 – 4.0 m
10	benzo(a)pyrene	soil	2 µg/g	640	0.3 – 4.0 m
10	phenanthrene	soil	6.8 µg/g	640	0.3 – 2.0 m
10	zinc	soil	158 µg/g	162	0 – 0.3 m
11	LEPHw	groundwater	680 µg/L	78	2.4 – 5.7 m
11	copper	soil	502 µg/g	333	0.3 – 1.5 m
11	copper	soil	158 µg/g	48	0.76 – 2.3 m
11	lead	soil	119 µg/g	83	0.1 – 1.5 m
11	tin	soil	71 µg/g	48	0.76 – 2.3 m
11	antimony	soil	99 µg/g	39	0.76 – 1.5 m
11	barium	soil	518 µg/g	39	0.76 – 1.5 m
11	cadmium	soil	2.7 µg/g	39	0.76 – 1.5 m
11	lead	soil	819 µg/g	39	0.76 – 1.5 m
11	zinc	soil	1410 µg/g	39	0.76 – 1.5 m
13	LEPHw	groundwater	680 µg/g	78	2.0 – 5.1 m
13	copper	soil	160 µg/g	51	0.76 – 1.5 m

Notes:

4.7 Offsite Migration

	Yes	No
Is there evidence that one or more substances has migrated or is likely to have migrated to a neighbouring site and is or is likely causing contamination of the neighbouring property?	☐	☒
Has any sampling occurred offsite for PCOCs in any media?	☐	☒
Have preferential pathways been assessed? (including assessment of all neighbouring underground utility rights-of-way)	☒	☐

If yes to the first question, complete the following:

There is evidence of ☐ historical, or ☐ current offsite transport of contaminants from the site in:

- ☐ groundwater;
- ☐ surface water;
- ☐ vapours; and/or
- ☐ other

Briefly describe the nature of and evidence for offsite migration (either known, suspected or potential)

The impacted offsite lands are categorized as:

- ☐ having a potable groundwater source;
- ☐ being aquatic habitat, as formally defined;
- ☐ agricultural lands
- ☐ residential or urban parklands
- ☐ commercial land
- ☐ industrial land

4.8 Investigation or Interpretation Issues to be Addressed

Identify any issues regarding investigations or interpretations if the PSI and DSI information may not satisfy the requirements of CSR Sections 58 and 59 and applicable protocols and guidance documents. Briefly describe how these deficiencies will be addressed (examples include destroyed wells, completion of detailed delineation following building demolition or other proposed work at a later stage of remediation).

Part 5: Remediation Summary

5.1 Remediation Reporting Summary

		Yes	No	n/a
Risk Assessment	Completed?	☒	☐	☐
	Includes quantitative human health and ecological risk assessment report information or screening level risk assessment per EMA, CSR and current applicable ministry protocols, guidelines, checklists?	☒	☐	☐
Remediation Plan	Completed?	☐	☒	☐
	Includes CSR specified information for a Remediation Plan *(see CSR S.1, 16, 47)* and current applicable ministry protocols, guidelines, checklists, etc.?	☐	☒	☐
Confirmation of Remediation	Completed?	☒	☐	☐
	Includes CSR specified information *(see CSR S.49)* and any current applicable ministry protocols, guidelines, checklists for COR reports?	☒	☐	☐
Other Reports	Completed?	☐	☐	☒
	According to other guidelines? *(Provide explanation in notes below. Indicate how reports assist understanding of conditions and / or remediation.)*	☐	☐	☒
Notes:				

If completed remediation reports are not adequate or if reports have a different scope than those listed above in accordance with the CSR complete Section 5.6 - Outstanding Remediation Issues.

5.2 Proposed or Completed Remedial Activities

(Describe all aspects of remediation, including regulatory actions and activities to comply with numerical and/or risk-based standards)

Regulatory

(Notification of Independent Remediation, Approval in Principle, Certificate of Compliance, Determination, Restrictive Covenant, etc.)

Notification of commencement and completion of Independent Remediation for the 2012 remediation work was sent to the MOE on February 25, 2012.
No record of a NIR was found for the 2010 remediation

If commitments or conditions to be met are included in an Approval in Principle issued for the site, list these conditions and identify how they were met though remedial activities.

Remediation to comply with numerical standards/criteria

(Excavation / disposal of soil, Treatment of soil; Treatment of groundwater, etc.)
52 tonnes of CL+ soil was disposed of off-site and 404 tonnes of hazardous waste soil was disposed of off-site. Soil was sent to at Envirogreen in Princeton (metals) or at Quantum in Delta (petroleum hydrocarbons)

Remediation to comply with risk-based standards

Are either of the following intended for use at the site, or have they been carried out?

	Intended		Carried Out	
Screening Level Risk Assessment	☐ Yes	☒ No	☐ Yes	☒ No
Quantitative Risk Assessment	☒ Yes	☐ No	☒ Yes	☐ No

If yes for any above, complete Section 5.5 (Summary of Residual Contamination)

Describe risk management / exposure reduction methods intended or implemented and indicate the status of any measures. *(e.g.,* Physical / engineering: *monitoring, capping or barriers to exposure;* Institutional: *registration of restrictive covenants, financial security, etc.)*

Floor slab of workshop is maintained in place as a barrier to petroleum hydrocarbon contamination under the slab. Service pits were filled in and capped with concrete

Soil with copper exceedances was buried on-site under 1.5 m of clean backfill

Residual insitu contamination at depth, in the oil house area, is under 4 m of backfill.
Engineered wetlands have been constructed to mitigate H2S migration from AEC 1 to aquatic receiving environment.
The RA assumed that the AEC with surface soil contamination (4, 5, 6, 10, 11, 13) will continue to be used for commercial-type activities (e.g., parking, buildings) with no vegetation present outside of the area the currently contains vegetation (i.e., the helipad area). Should vegetation be planted in other areas of AEC 4, 5, 6, 10, 11 and 13, it must be underlain by at least 1m of clean soil. Deep rooting vegetation is prohibited in areas with soil contamination at depths greater than 1m.
Health and safety planning is recommended prior to subsurface construction or utility work within AEC 5, 6 and 13 to mitigate exposures to soil vapours by such workers.

Provide a reference to signed and stamped design drawings provided by a professional engineer for works installed at site boundaries to prevent recontamination of a site.

Report #		Page #		or Appendix #	

5.3 Summary of Remediation Plan

	AEC #	Contaminant(s) of Concern	Remediated to the following standard *(proposed or completed)*				Remediation Schedule		
	(Use same #s as for APECs in Table above)		Numerical (Standard, guideline or criteria)	Back-ground (attach Protocol 4 or 9 approval if applicable)	Hazardous Waste standard	Risk-based	Remediation Complete? Yes	Remediation Complete? No	Proposed or Actual completion date
Soil	3	HEPH				✓	✓		May 2012
	4	LEPH				✓	✓		
	4	HEPH				✓	✓		
	4	lead				✓	✓		
	4	zinc				✓	✓		
	5	HEPH				✓	✓		
	5	copper	✓ - RL/PL				✓		
	5	lead	✓ - RL/PL				✓		

	AEC #	Contaminant(s) of Concern	Remediated to the following standard *(proposed or completed)*				Remediation Schedule		
	(Use same #s as for APECs in Table above)		Numerical (Standard, guideline or criteria)	Back-ground (attach Protocol 4 or 9 approval if applicable)	Hazardous Waste standard	Risk-based	Remediation Complete? Yes	Remediation Complete? No	Proposed or Actual completion date
	5	tin				✓	✓		
	5	zinc				✓	✓		
	6	LEPH				✓	✓		
	6	HEPH				✓	✓		
	6	benzo(a)anthracene				✓	✓		
	6	benzo(b)fluoranthene				✓	✓		
	6	benzo(k)fluoranthene				✓	✓		
	6	benzo(a)pyrene				✓	✓		
	6	indeno(1,2,3-cd)pyrene				✓	✓		
	10	LEPH	✓ - RL/PL			✓	✓		
Soil	10	HEPH	✓ - RL/PL			✓	✓		
	10	ethylbenzene	✓ - RL/PL				✓		
	10	xylenes	✓ - RL/PL				✓		May 2012
	10	VPH	✓ - RL/PL			✓	✓		
	10	benzo(a)pyrene	✓ - RL/PL			✓	✓		
	10	zinc	✓ - RL/PL			✓	✓		
	11	copper	✓ - RL/PL			✓	✓		
	11	copper	✓ - RL/PL			✓	✓		
	11	lead	✓ - RL/PL			✓	✓		
	11	tin	✓ - RL/PL			✓	✓		
	11	antimony	✓ - RL/PL			✓	✓		
	11	barium	✓ - RL/PL				✓		
	11	cadmium	✓ - RL/PL				✓		
	11	lead	✓ - RL/PL				✓		
	11	zinc	✓ - RL/PL			✓	✓		
	13	copper	✓ - RL/PL			✓	✓		
Groundwater	3	non-chlorinated phenols				✓	✓		May 2012
	1	sulphide	✓ - AW				✓		November 2012
	1	chloride	✓ - AW				✓		
	6	LEPHw				✓	✓		
	6	naphthalene				✓	✓		May 2012
	11	LEPHw					✓		
	13	LEPHw				✓	✓		
Sediment	N/A								
fac e Wa	1	sulphide	✓ - BC Water Quality Guidelines				✓		November 2012
	1	chloride					✓		

	AEC #	Contaminant(s) of Concern	Remediated to the following standard (proposed or completed)				Remediation Schedule		
	(Use same #s as for APECs in Table above)		Numerical (Standard, guideline or criteria)	Back-ground (attach Protocol 4 or 9 approval if applicable)	Hazardous Waste standard	Risk-based	Remediation Complete? Yes	Remediation Complete? No	Proposed or Actual completion date
Soil Vapour	4	benzene	✓ - RL/PL						May 2012

In the AEC column, specify as N/ A (not applicable) if remediation or assessment is not required in this environmental medium.

5.4 Summary of Contaminant Treatment or Removal

(Identify and describe all contamination removed from or treated on-site. Ensure section 6.2 is completed if no CSRA is required or only required for a portion of soil removed)

Provide references to figure(s) showing the lateral and vertical extent of any treated or removed contamination. Confirmatory sample locations and corresponding analytical results shall be shown on each figure and in tabular form with reference to applicable standards:

Environmental Medium	Soil	Report #	2	Figure #	14, 15, 16	Page #	93
Environmental Medium	Groundwater	Report #	1	Figure #	15	Page #	
Environmental Medium		Report #		Figure #		Page #	

AEC / APEC # (Use same #s as for APECS in Table above)	Contaminant(s) of Concern	Medium (e.g., soil, groundwater, sediment, surface water, vapour, other)	Material Treated/Removed: Volume (m^3 or L)	Material Treated/Removed: Disposal Location (indicate if treated on-site)
10	LEPH, HEPH, ethylbenzene, xylenes, VPH, benzo(a)pyrene	Soil	2560 m^3	Quantum, Delta, BC
10	Antimony, barium, cadmium, copper, lead, zinc	Soil	620 m^3	Envirogreen, Princeton, BC
5	lead	Soil	38 m^3	Envirogreen, Princeton, BC
13	Antimony, copper, lead, zinc	Soil	320 m^3	Buried on-site at depth of >1.5 m at former oil house excavation
1	Sulphide and Chlorides	Groundwater	n/a	Engineered Wetland on-Site

5.5 Summary of Residual Contamination after Remediation

(Identify and describe all contamination that exceeds CSR numerical standards, after the remediation described above has been implemented.)

AEC / APEC #	Contaminant of Concern	Medium (e.g., soil, groundwater, sediment, surface water, vapour, other)	Maximum Measured Concentration (indicate units)	Extent of Contamination	
				Area (m^2)	Depth Range (m)
3	non-chlorinated phenols	groundwater	413 µg/L	450	n/a
3	HEPH	Soil	3,270 µg/g	413	1.5 – 2.3 m
4	LEPH	Soil	1,730 µg/g	487	0.3 – 1.5m
4	HEPH	Soil	1,780 µg/g	487	0.3 – 1.5m
4	Lead	Soil	223 µg/g	20	0 – 0.5 m
4	Zinc	Soil	201 µg/g	285	1.5 – 3.5 m
4	Zinc	Soil	186 µg/g	320	0.76 – 1.5 m
4	benzene	Vapour	2.6 µg/m^3	384	
5	HEPH	Soil	1300 µg/g	318	0 – 1.5 m
5	Copper	Soil	185 µg/g	216	0 - 2.9 m
5	Lead	Soil	553 µg/g	206	0 - 2.9 m
5	Tin	Soil	625 µg/g	190	0.76 – 3.5 m
5	Zinc	Soil	898 µg/g	190	0.76 – 3.5 m
6	LEPH	Soil	2,900 µg/g	316	1.52 – 2.6 m
6	LEPHw	groundwater	1,300 µg/L	190	2.5 – 7.1 m
6	naphthalene	groundwater	22 µg/L	190	2.5 – 7.1 m
6	HEPH	Soil	1300 µg/g	316	0.76 – 2.6 m
6	benzo(a)anthracene	Soil	2.5 µg/g	316	0.76 – 2.6 m
6	benzo(b)fluoranthene	Soil	3.3 µg/g	316	0.76 – 2.6 m
6	benzo(k)fluoranthene	Soil	1.1 µg/g	316	0.76 – 2.6 m
6	benzo(a)pyrene	Soil	2.2 µg/g	316	0.76 – 2.6 m
6	indeno(1,2,3-cd)pyrene	Soil	1.6 µg/g	316	0.76 – 2.6 m
10	LEPH	Soil	5,100 µg/g	802	1.2 – 4.0 m
10	HEPH	Soil	13,800 µg/g	802	1.2 – 4.0 m
10	VPH	Soil	450 µg/g	640	1.2 – 4.0 m
10	antimony	Soil	99 µg/g	128	1.5 – 4.0 m
10	cadmium	Soil	2.7 µg/g	128	1.5 – 4.0 m
10	Lead	Soil	819 µg/g	128	1.5 – 4.0 m
10	Zinc	Soil	158 µg/g	162	0 – 0.3 m
11	LEPHw	groundwater	680 µg/L	78	2.4 – 5.7 m
11	Copper	Soil	535 µg/g	333	0.3 – 1.5 m
11	Copper	Soil	158 µg/g	48	0.76 – 2.3 m
11	Lead	Soil	119 µg/g	83	0.1 – 1.5 m
11	Tin	Soil	71 µg/g	48	0.76 – 2.3 m
13	LEPHw	groundwater	680 µg/g	78	2.0 – 5.1 m
13	Copper	Soil	375 µg/g	51	0.76 – 1.5 m

5.6 Remediation Issues

Identify remaining issues if the remediation plan, confirmation of remediation report or risk assessment report does not include CSR specified information and current applicable ministry protocols, guidelines, checklists, etc. for these documents..

Part 6: Summary of Soil Relocation

6.1 Relocation with a Contaminated Soil Relocation Agreement

Source Site

(Soil to be relocated under the CSRA (from Table 4.6). Investigation information may be limited to the soil that is the subject of the relocation agreement)

APEC # *(Use same #s as for APECS in Table above)*	**Contaminant of Concern** *(List separately)*	**Classification of the soil to be Relocated** *(</>AL, PL, RL, CL, IL; Column II, III, IV)*		**Volume m^3**
		Schedule 4, 5 or 10	**Schedule 7**	

Receiving Site

Soil to be relocated has been adequately characterized?				☐ Yes	☐No
Total Volume of soil to be relocated?					m^3
Applicable CSR Land Use at receiving site	AL ☐	PL ☐	RL ☐	CL ☐	IL ☐

Contaminant *(as indicated in CSR Schedule 7)*	**Maximum Contaminant Concentration in soil to be relocated (mg/kg)**	**Applicable CSR Schedule 4 or 5 standard at receiving site (mg/kg)**

Sufficient data on receiving site?	☐ Yes	☐No

(Ensure to assess any modifying factors for the receiving site soils such as soil pH)

Conditions pertaining to relocation (CSR, Sec. 44):

Will the source and receiving municipality be notified before soil is relocated?	☐ Yes	☐No
Will at least 4 business days be allowed to pass before soil is relocated?	☐ Yes	☐No

6.2 Relocation without a Relocation Agreement

Other soil relocation not requiring a Contaminated Soil Relocation Agreement (CSRA):

Has or will contaminated soil be relocated without a CSRA?	☒ Yes	☐ No
Do exemptions apply? *(indicated below; see CSR Sec. 41)*		
Relocation of contaminated soil on the site at which the contaminated soil originates?	☒ Yes	☐ No
Relocation of contaminated soil which is contaminated due only to the presence of the local background concentration?	☐ Yes	☒ No
Relocation of contaminated soil within an area subject to a wide area remediation plan?	☐ Yes	☒ No
Relocation of contaminated soil originating from emergency cleanup of a spill?	☐ Yes	☒ No
Relocation of soil to an authorized hazardous waste storage or treatment facility? If yes, provide BC Generator Registration #_____	☐ Yes	☒ No
Relocation of contaminated soil to a destination outside of British Columbia?	☐ Yes	☒ No
Relocation of contaminated soil from a specific site not exceeding 5 cubic metres in volume?	☒ Yes	☐ No
Relocation of contaminated soil to federal property?	☐ Yes	☒ No
Relocation to an authorized landfill that is exempt from a CSRA?	☒ Yes	☐ No

Part 7: Recommendation of Approved Professional(s)

7.1 Regulatory Instrument and Summary Recommendation

Based on the detailed technical information available for the site, as summarized in this Summary of Site Condition, I, *Lori C. Larsen, P.Ag and Scott Steer, R.P.Bio.*, recommend that the following instrument be issued for the Subject Site.

- ☐ A Determination under section 44 of EMA
- ☐ An AiP under section 53 (1) of EMA
- ☒ A CoC under section 53(3) of EMA
- ☐ A CSRA under section 55(2) of EMA
- ☐ Other (specify)

Although I understand that the basis of such recommendations should only be formally evaluated by reference to detailed technical guidance, the primary basis of this recommendation or these recommendations is as follows:

- A Preliminary Site Investigation addressing all identified areas of potential environmental concern (APECs) and potential contaminants of concern (PCOCs) was completed. One or more substances were identified at concentrations exceeding applicable standards or criteria in CSR Schedules 4, 5, 6, 7 10, or 11, or criteria in Schedule 9.
- A Detailed Site Investigation addressing the locations and extent of all identified areas of environmental concern (AECs) and contaminants of concern was completed and forms the basis of a risk assessment and was the basis for remediation of the site..
- When this Summary of Site Condition was prepared all contaminants and their respective locations and extent as identified in investigation and risk assessment reports had been remediated to CSR numerical concentration or risk-based standards or criteria (sediments) or both.

7.2 Substances Remediated and Standards or Criteria

Contaminants have been remediated to comply with standards or criteria listed in the following table:

(If the site required remediation and has been remediated.)

	Substances Remediated for Each Type of Standard / Criteria	
	Numerical Standards (Criteria for Sediments)	**Risk based Standards (Criteria for Sediments)**
Soil	Antimony	Antimony
	Barium	Tin
	Cadmium	Copper
	Tin	Lead
	Copper	Zinc
	Lead	VPHs
	Zinc	LEPH
	VPHs	HEPH
	LEPH	Ethylbenzene
	HEPH	Xylene
	Ethylbenzene	Benzo[a]anthracene,
	Xylene	benzo[b]fluoranthene,
	benzo[a]pyrene (B[A]P)	benzo[k]fluoranthene
		indeno[1,2,3-c,d]pyrene
		phenanthrene
		benzo[a]pyrene (B[A]P)
Water	Chloride	non-chlorinated phenols
	Sulphide (as H_2S)	LEPHw
		naphthalene
Sediments		
Soil Vapour	Benzene	
Other		

Use specific compound names as listed in the Contaminated Sites Regulation schedules

The Director may accept the recommendations of an Approved Professional(s) involved in the review and submission of investigation, risk assessment or remediation reports based in part on the understanding that:

- As of November 23, 2012 the date of signing of this report the Approved Professional, or Approved Professionals if more than one, is/are member(s) in good standing of the Roster of Approved Professionals, as maintained by the ministry, and member(s) of the Contaminated Sites Approved Professionals Society (CSAP Society);
- The Approved Professionals signing this Summary of Site Condition have reviewed Table 1, Protocol 6 for Contaminated Sites (*Eligibility of Applications for Review by Approved Professionals*) and confirm that the *Application for Contaminated Sites Services* may be processed in the manner for non-high risk sites under the Roster of Approved Professionals process;

- That the reviewer has no obligation to undertake any inquiry into the validity, accuracy or precision of what is reported in the documents reviewed, beyond that which there is reasonable cause to believe that there could be errors or oversights in those reports;
- The subject site has been satisfactorily investigated for all areas of potential environmental concern and potential contaminants of concern to determine the lateral and vertical extents of contamination with due regard to the *EMA*, the CSR, and the HWR;
- The submitted documentation meets the requirements of Sections 1, 49, 58 and 59 of the CSR;
- The confirmation of remediation report meets the requirements of section 49(2) of the CSR;

The opinions, advice and recommendations expressed in this Summary of Site Condition are made in accordance with generally accepted principles and practices as recognized by members of the applicable profession or discipline practising at the same time and in the same or similar locations. This Summary of Site Condition does not provide a legal opinion or guarantee regarding compliance with applicable laws.

Name(s) of Approved Professional(s):	Signature(s) of Approved Professional(s):	Date:
Lori C. Larsen, P.Ag.		Nov 23/12
Scott A. Steer R.P. Bio.		Nov 23/12

7.3 Arm's Length Review

There may have been an arm's length review of one or more of the following recommendations to the Director of Waste Management:

1. Making a recommendation to a Director in support of an application for an Approval in Principle based on remediation to numerical standards or a screening level risk assessment where there is offsite migration at the site.
2. Making a recommendation to a director in support of an application for an Approval in Principle based on a risk assessment (other than a screening level risk assessment) and remediation to risk-based standards
3. Making a recommendation to a Director in support of an application for a Certificate of Compliance based on remediation to numerical standards or a screening level risk assessment where there is offsite migration at the site.

4. Making a recommendation to a Director in support of an application for a Certificate of Compliance based on a risk assessment (other than a screening level risk assessment) and remediation to risk-based standards
5. Making a recommendation to a Director in support of an application for a Contaminated Soil Relocation Agreement based on a screening level risk assessment for the parcel at which the soil is to be deposited where there is offsite migration at the source site
6. Making a recommendation to a Director in support of an application for a Contaminated Soil Relocation Agreement based on a risk assessment (other than a screening level risk assessment) for the parcel at which the soil is to be deposited
7. Making a recommendation to a Director in support of any other application based on risk assessment or risk management (other than a screening level risk assessment) not otherwise described in any other row in this list, as required under a protocol signed by a Director.

If this is the case please have the Approved Professional who carried out the arm's length review to sign below, specifying the type of arm's length review done for the site.

Type of Arm's Length Review (Insert number from list)	Name(s) of Approved Professional(s):	Signature(s) of Approved Professional(s):	Date:
4	Scott A. Steer R.P. Bio.		Nov 23/12

Part 8: Statement of Site Owner / Agent / Lessee

8.1 Offsite Migration Notification – N/A

If it is known that one or more substances has migrated or is likely to have migrated to a neighbouring site and is or is likely causing contamination of the neighbouring site, have notifications been given?

(See CSR Sec. 57 and 60.1 for requirements)

	Yes	No
Have owners of impacted offsite properties been formally notified?	☐	☐
Has the ministry been formally notified?	☐	☐

8.2 Confirmations by Owner / Agent / Lessee Regarding Approved Professional

This is to acknowledge that as ~~<the owner~~ MG / as the agent on behalf of the owner / ~~lessee>~~ MG *(strike out and initial that which does not apply – if signing as the agent of the owner or lessee, written consent from the owner or lessee authorizing signature of this Summary of Site Condition must be attached)* of the site I have engaged *Lori C. Larsen, P.Ag. and Scott Steer, R.P. Bio.* as the Approved Professional(s) to review site investigation, risk assessment and remediation reports and to make submission and application with recommendations, if applicable, for the regulatory instrument(s) as indicated in this Summary of Site Condition.

I agree to comply with any requirements on the site regarding monitoring and maintenance of works as documented in Schedule B of the draft contaminated sites legal instrument.

I accept that if a risk assessment has been applied at the site, that the risk assessment is only valid as long as conditions at the site do not change.

I have undertaken reasonable inquiry into the previous ownership and uses of the property and to the best of my knowledge I have provided to the Approved Professional, information relevant to the investigation and remediation of the environmental condition of the site, in the preparation of this document.

I acknowledge that this Summary of Site Condition becomes a public document after it has been received and acknowledged by the Director of Waste Management. Any party intending to purchase, lease, take a security interest in, or occupy the site may review this document and any supporting documents to satisfy themselves with respect to the environmental condition of the site, and the extent of responsibility and liability that may arise from taking ownership, taking a security interest, or occupying the site.

I have made no modifications to this document except as allowed by the form.

Name:	*Michael Geraghty (Keystone Environmental Ltd.)*
Address:	*Suite 320 – 4400 Dominion Street, Burnaby, BC V5G4G3*
Signature:	
Date:	*November 23, 2012*

1